Enabling Near-Zero Cost Object Detection in Remote Sensing Imagery via Progressive Self-Training

Xiang Zhang, Xiangteng Jiang, Qiyao Hu, Hangzai Luo, Sheng Zhong, Lei Tang, Jinye Peng, and Jianping Fan

***Abstract*— Deep learning-based object detection models rely heavily on large-scale and precise annotations for training. However, manually annotating bounding-box annotations for such data is both time-consuming and costly, especially when dealing with high-resolution satellite imagery containing densely packed small-sized objects. To alleviate the burden of manual annotation, we propose a simple yet effective approach, called progressive self-training object detection (PSTDet), to enable accurate object detection in remote sensing imagery without relying on manual annotations. Our PSTDet framework consists of two main components: initial pseudo label generation (IPLG) and progressive self-training with relabeling (PST-R). In IPLG, we leverage unsupervised image clustering, unsupervised instance detection, and geometric constraints to automatically generate high-quality bounding-box annotations for the initial training dataset. This innovative approach significantly reduces the time and expense associated with data annotation, laying a solid foundation for the subsequent progressive self-training stage. The annotations produced by IPLG serve as the training data for PST-R, which enhances the detector and pseudo labels through progressive self-training and our proposed noisy pseudo label filtering strategy (NPLFilter). Our NPLFilter purifies the quality of pseudo labels by integrating geometric constraints, prior knowledge, and category-adaptive thresholds. Experimental results demonstrate that our method achieves significant performance improvement on challenging NWPU VHR-10.v2 and DIOR datasets. Notably, our method far outperforms state-of-the-art weakly supervised methods and compares favorably with fully supervised methods.**

***Index Terms*— Deep networks, object detection, remote sensing images, self-training.**

I. Introduction

DEEP learning has demonstrated impressive performance in object detection [1], [2], [3], change detection [4], [5], semantic segmentation [6], [7], and other areas. Among these, remote sensing object detection not only aids in monitoring natural resource utilization and urban development but also serves as a crucial tool in disaster monitoring and environmental protection efforts, thereby garnering widespread attention. Accurate object detection heavily relies on large-scale training data with bounding-box annotations. However, the complexity and diversity of objects in remote sensing images, combined with factors such as diverse object scales and densely distributed objects, present significant challenges for manual annotation of bounding boxes (see Fig. 1).



(a) (b)

Fig. 1. In remote sensing images, (a) objects often appear at very different scales, as illustrated by the bounding boxes and (b) objects are often densely arranged.

In recent years, there has been a growing interest in semi-supervised object detection (SSOD) as a potential solution to alleviate the burden of data annotation. SSOD [8], [9], [10] leverages a combination of a limited annotated dataset and a large-scale unlabeled dataset to optimize the model and reduce reliance on labeled data. However, it is important to note that SSOD assumes that the labeled and unlabeled data are sampled from the same underlying distribution. In real-world scenarios, this assumption may not consistently hold. Additionally, the performance of SSOD not only depends on manual annotation of a subset of the data but also heavily relies on the quality of the labeled data. To further alleviate the burden of data annotation, a significant amount of attention has been directed toward image-level weakly supervised object detection (WSOD) techniques [11], [12], [13], [14], [15]. These approaches rely on supervision at the image level to identify object regions, eliminating the need for precise bounding-box annotations. Despite notable advancements in WSOD techniques, their performance still lags behind that

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Fig. 2. Unreliable bounding-box annotations of training instances.

of fully supervised object detection (FSOD) counterparts. This discrepancy primarily stems from the fact that weakly supervised learning overlooks the intrinsic properties captured by bounding-box annotations, resulting in imprecise learned detection knowledge and challenges in accurately localizing objects. Furthermore, typical feature representations learned from weakly supervised settings only capture salient parts of representative objects, deviating from the desired detection results (as shown in Fig. 2).

To overcome the aforementioned challenges, we introduce a novel approach called progressive self-training object detection (PSTDet). PSTDet is designed to gradually acquire detection knowledge by leveraging both initial training data and unlabeled data. The framework of PSTDet consists of two core components: initial pseudo label generation (IPLG) and progressive self-training with relabeling (PST-R). To address the requirements of SSOD for annotated data and overcome the issue of WSOD neglecting the intrinsic characteristics captured by bounding-box annotations, we design the IPLG component to automatically generate bounding-box annotations for large-scale unlabeled initial training data. Furthermore, our PST-R component is designed to address the distribution gap between the initial training data and the target domain data, thereby resulting in improved overall detection performance. Within the IPLG component, we synergistically integrate unsupervised clustering algorithms, unsupervised instance detection paradigms, and geometric constraint methodologies to facilitate the automated generation of bounding-box annotations on the initial training data. This process leverages advanced techniques to produce high-quality initial annotations that serve as the training data for our proposed PST-R. The PST-R iteratively optimizes the detector and enhances the quality of pseudo labels by leveraging both the initial training data with bounding-box annotations and unlabeled data. To mitigate the negative impact of noisy pseudo labels, we introduce noisy pseudo label filtering strategy (NPLFilter), a label filtering strategy that enhances the quality of pseudo labels by integrating geometric constraints, prior knowledge, and category-adaptive thresholds.

Fig. 3 illustrates a comparison of our PSTDet with fully supervised, semi-supervised, and WSOD methods. Traditional approaches often rely on bounding-box annotations or image-level labels within the target domain. In contrast, our PSTDet eliminates the need for labels (i.e., image-level labels and bounding-box annotations) in the target domain, relying instead on assigning category labels to files obtained through unsupervised clustering during the IPLG phase. This process enables the training of a detector capable of identifying specific objects like vehicles or ships. For example, if we

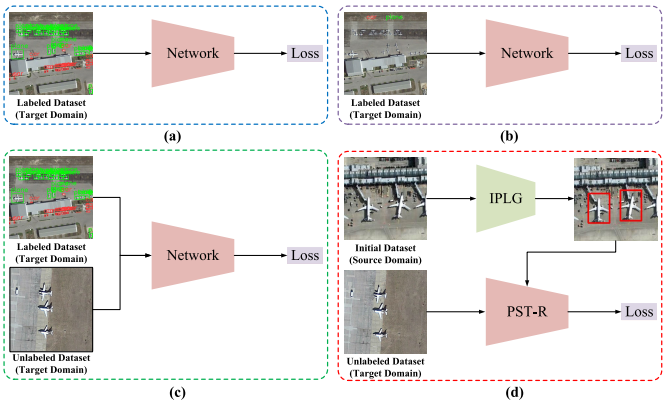


Fig. 3. Comparison between (a) supervised learning, (b) weakly supervised learning, (c) semi-supervised learning, and (d) our proposed PSTDet.

obtain five folders through unsupervised clustering, manual observation and labeling of these folders with category labels are required. In contrast to annotating large-scale training data with image-level or bounding-box annotations, the resource consumption of our approach is negligible, leading us to term it "near-zero cost." As a result, our PSTDet combines the strengths of semi-supervised (strong performance) and weakly supervised (inexpensive labeling) object detection techniques, effectively addressing their inherent limitations such as expensive labeling and poor performance.

Our main contributions are summarized below as follows.

1) We propose PSTDet, an automated framework that facilitates object detection in remote sensing imagery by eliminating the need for manual bounding-box annotation.
2) We present the IPLG strategy, which autonomously generates accurate annotations for initial unlabeled training data in an unsupervised manner. Additionally, we develop a PST-R to iteratively optimize the model and refine pseudo labels.
3) We introduce NPLFilter, an effective strategy that enhances model optimization during iterative self-training by eliminating noisy pseudo labels.
4) Our approach beats state-of-the-art WSOD performance on both the NWPU VHR-10.v2 [16] and DIOR [17] datasets by significant margins. Furthermore, our method even surpasses most existing fully supervised methods.

II. Related Work

A. Weakly Supervised Learning

With the booming development of deep learning, object detection has been extensively studied for remote sensing images. For example, the ORSIm detector [18] is an efficient object detection framework that leverages rotation-invariant channel features and learning methods to capture important and meaningful features for detecting objects in optical remote sensing imagery. Wu et al. [19] customized a UIU-Net framework for multilevel and multiscale feature aggregation to achieve high-precision detection of small objects in infrared imagery. Nevertheless, the efficacy of these methods heavily relies on the existence of extensively annotated datasets. The

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manual annotation of accurate bounding boxes at a large scale is not only laborious and costly but also presents challenges in maintaining consistent labeling quality. Recently, weakly supervised learning has been widely applied in the field of remote sensing object detection to overcome the challenge of insufficient annotated data, and significant progress has been made. In particular, some WSOD methods only use image-level labels to guide the training of detectors. These methods exploit easily obtainable classification labels to infer spatial localization cues for object detection, eliminating the necessity for bounding-box annotations. Feng et al. [20] proposed a novel triple context-aware network (TCANet) to learn complementary and discriminative visual patterns, aiming to enhance the performance of WSOD in remote sensing imagery. Yao et al. [21] designed a dynamic curriculum learning strategy that provides the detector with images of increasing difficulty by evaluating the difficulty of objects in the images. This approach aims to progressively enhance the performance of the detector. Guo et al. [13] proposed a method that integrates the transformer with weakly supervised learning to enhance the representational capacity of the detector. Cheng et al. [11] introduced a self-guided proposal generation method that can be easily integrated into a unified framework with most WSOD methods. Tan et al. [14] developed a dynamic informative proposal-based iterative WSOD training approach, which gradually improves the quality of information proposals through the use of the information proposal reconstruction (IPR) method, proposal-based contrastive learning (PBCL) technique, and pseudo label learning-based multistage training procedure (PLMS). Qian et al. [15] tackled the problem of WSOD focusing on local discriminative regions rather than the entire object by introducing the multiscale image splitting-based feature enhancement (MSFE) module. Additionally, they proposed the instance difficulty-aware training (IDAT) strategy to address the imbalance between difficult and easy instances. This type of algorithm overlooks the intrinsic properties of bounding-box annotations, hindering accurate perception of object location information. While weakly supervised methods reduce annotated data costs, they often exhibit unstable performance and necessitate more complex algorithms compared to fully supervised methods. In this article, we introduce an unsupervised strategy aimed at generating high-quality bounding-box annotations, which can significantly alleviate the time-consuming burden of manual annotation.

B. Pseudo Label Learning

The pseudo label learning technique enhances model training by providing additional supervisory information through the generation of pseudo labels using diverse and sophisticated strategies. This technique has gained significant attention and acclaim in semi-supervised learning, especially in SSOD. Yang et al. [22] proposed an interactive self-training technique for SSOD that effectively combines detection outcomes from various iterations through the utilization of nonmaximum suppression. Furthermore, multiple detection heads are employed to enhance the precision of pseudo labels, consequently improving the overall accuracy of the detection model. Li et al. [23] introduced confidence-aware pseudo labels tailored for object detection to accurately estimate the classification and localization quality of derived pseudo labels. Chen et al. [8] proposed a novel DenSe Learning (DSL) method to enhance the utilization of large-scale unlabeled data in SSOD. The method assigns fine-grained pseudo labels to each pixel using an adaptive filtering (AF) strategy and introduces aggregated teachers (AT) to improve the stability and quality of estimated pseudo labels. Vandeghen et al. [24] developed an adaptive self-training for object detection (ASTOD), which selects reliable pseudo labels by adaptively examining the score histogram. Shen et al. [9] introduced a prospective consistent teacher framework for object detection, which enhances the performance of SSOD using weighted dense proposal learning (WDPL), mean-consistency-based proposal pruning (MCPP), and EM-based fitting policy (EFP). However, semi-supervised learning still depends on the manually labeled data and assumes that unlabeled data and labeled data have the same domain distribution. Our approach distinguishes itself by introducing our proposed IPLG and PST-R techniques. First, IPLG autonomously produces precise bounding-box annotations for the initial training data, providing a robust foundation for the training process. Subsequently, our PST-R method refines pseudo labels, progressively enhancing their quality. These improved pseudo labels are then utilized to further train the model, enhancing its performance.

C. Pseudo Label Filtering

Deep models are often affected by noisy labels during the training process, which inevitably reduces the robustness of the model. Most existing self-training object detection methods [25], [26], [27] mainly adopt manually set confidence thresholds to filter out low-quality pseudo labels. In particular, Sohn et al. [28] designed a semi-supervised learning approach based on self-training and reinforcement-driven consistency regularization. They utilize teacher-generated candidate labels and incorporate pseudo labels with confidence scores higher than the threshold of 0.9 as part of the training data in the next round. Li et al. [29] introduced a selective self-supervised self-training method for object detection. They filter out pseudo labels using a confidence threshold of 0.7 and improve the pseudo labeling step through a selective network. Li et al. [30] constructed a teacher–student framework called adaptive teacher, which leverages domain adversarial learning and weak–strong data augmentation to address domain discrepancies. In terms of label purification, they also employed a confidence threshold of 0.8 to filter out noisy pseudo labels. Although these methods have played a role in performance improvement, the selection of threshold values remains a technical challenge. A higher threshold value leads to more reliable pseudo labels but may result in insufficient training data. On the other hand, a lower threshold value ensures an adequate amount of training data but introduces excessive noise. In this article, we present NPLFilter, a novel approach that leverages geometric constraints, prior knowledge, and category-adaptive thresholds to improve the quality of pseudo labels. By incor-

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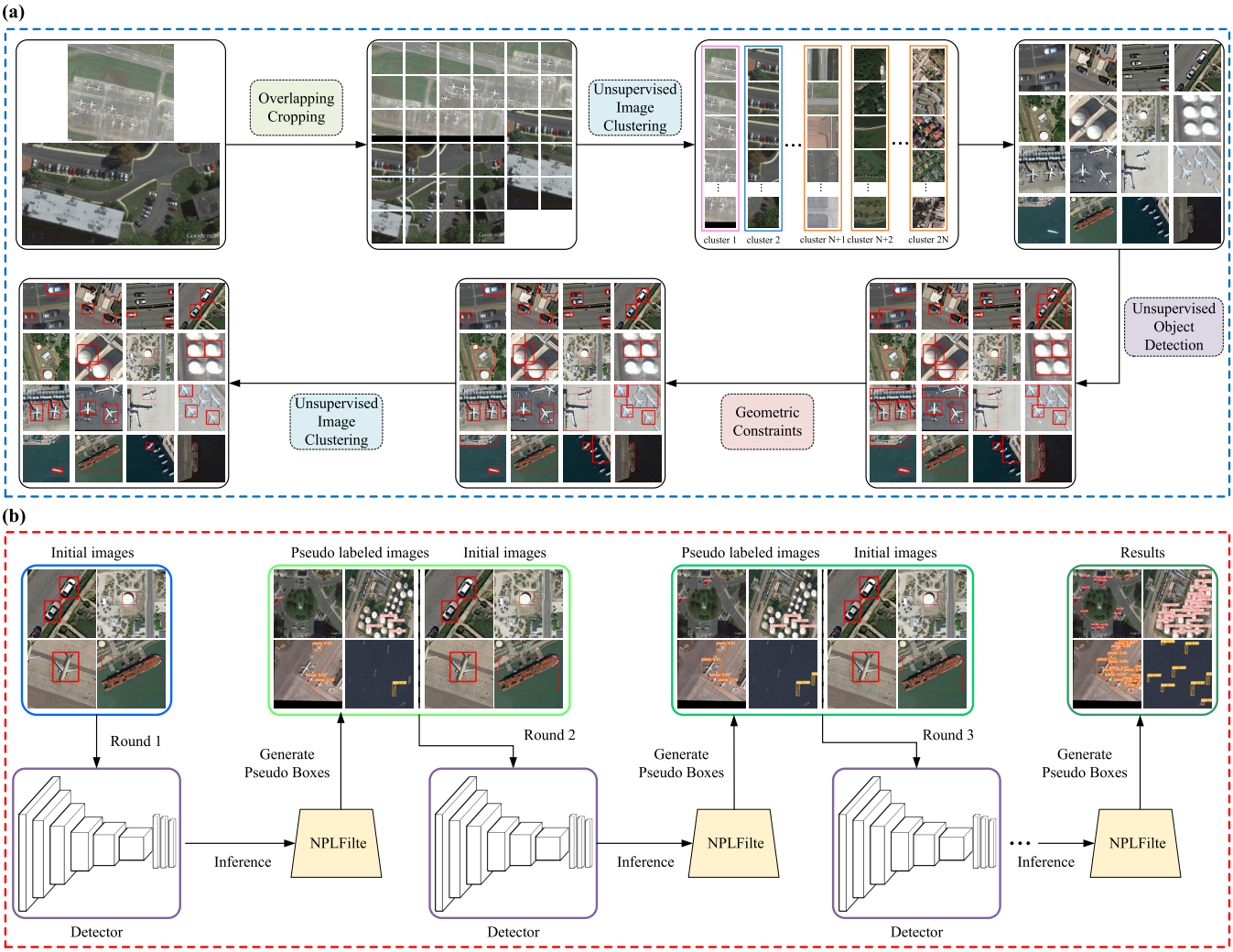


Fig. 4. Pipeline of the proposed PSTDet framework. NPLFilter represents the proposed pseudo box filtering strategy, which is used to filter the unreliable labels of training instances. (a) IPLG. (b) PST-R.

porating these techniques, our method enhances the model optimization throughout the progressive self-training process.

III. Methodology

A. Overview of the Proposed Method

The high performance of object detection relies heavily on large-scale training data with accurate bounding-box annotations, posing a significant burden on annotation tasks. Achieving high-precision object detection while alleviating the heavy workload of data annotation presents a highly challenging endeavor. To this end, we introduce PSTDet, as depicted in Fig. 4. Our PSTDet begins with the utilization of IPLG to generate precise bounding-box annotations for the initial training data. Subsequently, PST-R and NPLFilter are employed to iteratively optimize the pseudo labels and detectors. IPLG plays a crucial role in enhancing the model's performance by providing accurate initial bounding-box annotations. PST-R enables the refinement of pseudo labels across multiple iterations, ensuring the utilization of high-quality annotations during model training. Moreover, NPLFilter aids in eliminating unreliable annotations, further enhancing the accuracy of the model.

B. Initial Pseudo Label Generation

The workflow of the IPLG is depicted in Algorithm 1. The specific details of the related algorithms are expounded upon in Sections III-B1–III-B4.

1) Image Acquisition and Preprocessing: During the IPLG phase, an important task is to acquire and process remote sensing images. To fulfill this requirement, it is necessary to collect publicly available large-scale remote sensing datasets, such as RSOD [33], LEVIR [34], NWPU VHR-10.v2 [16], DIOR [17], and UCAS-AOD [35], among others. These datasets cover remote sensing images from different regions, periods, and resolutions, providing rich semantic information and diverse target categories. Then, an overlapping cropping operation is used to divide the images in datasets other than the DIOR dataset into multiple patches. This is done because the object instances in the DIOR dataset have large sizes, and overlapping clipping cannot guarantee the integrity of the objects. Compared to the entire remote sensing image, the patches

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Algorithm 1 Framework of IPLG

Require: Remote sensing images: D^o.

Ensure: Bounding-box Annotations.

1: Collect remote sensing images from multiple publicly available remote sensing datasets to construct a large-scale remote sensing image dataset D^o, and divide the images in this dataset into multiple patches using the overlapping cropping operation;
2: Apply the unsupervised clustering algorithm (i.e., K-Means++ [31]) to cluster these image patches into $2N$ categories, where N represents the number of categories in the dataset;
3: Remove image patches that do not include the vehicle, storage tank, airplane, and ship classes;
4: Use the unsupervised instance detection model (i.e., CutLER [32]) to generate high-quality bounding-box annotations for clustered image patches;
5: Employ geometric constraints and unsupervised image clustering to purify pseudo labels, and the remaining pseudo labels and their corresponding image patches constitute the initial training data.

obtained through cropping have simpler backgrounds, which is beneficial for subsequent unsupervised image clustering and instance detection.

2) Unsupervised Image Clustering: In this study, ResNet34 [36] is used for feature extraction, and K-Means++ [31] is adopted for image clustering. Specifically, after obtaining the cropped patches, ResNet34 is utilized to extract features from these patches. Subsequently, the extracted features are fed into the K-Means++ to cluster the patches into $2N$ categories, where N represents the number of categories in the dataset. Although the cropped dataset background is simpler and facilitates subsequent unsupervised image clustering and unsupervised instance detection, it inevitably results in additional categories and different background images. For example, overlapping cropping of the ROSD dataset will produce objects such as airstrips, grasses, buildings, and so on. Therefore, we will cluster patches into $2N$ categories instead of N categories after inputting the extracted features into K-Means++. In this article, we mainly study the detection of vehicles, storage tanks, airplanes, and ships. Therefore, we directly discard the clustered images of other categories.

3) Unsupervised Object Detection: To obtain high-quality bounding-box annotations of the unlabeled data for training, we employ an unsupervised object detection model called CutLER [32]. CutLER leverages the intrinsic properties of the self-supervised model to discover objects and enhances their features in an unsupervised manner. Since cropped patches are less disturbed by background information, CutLER can more easily obtain bounding-box annotations for instances in an image.

4) Initial Noisy Pseudo Label Filtering: The pseudo labels generated from the aforementioned strategy may not always contain accurate labels. Training a detection model with data that includes low-quality pseudo labels can significantly degrade the performance of the model. To address this issue, we integrate geometric constraints and unsupervised image clustering to filter out noisy pseudo labels. By observing the relative position and size of the objects, reasonable ranges of positions can be inferred based on common sense and prior knowledge. The shape, size, and position information of the target is then used to filter out noisy pseudo labels, which are extremely unreliable. Specific constraints are defined as follows:

$$\left(\frac{w}{W} < T_{\text{size}}\right) \wedge \left(\frac{h}{H} < T_{\text{size}}\right) \tag{1}$$

$$\left|\frac{w}{h} - 1\right| < T_{\text{shape}} \tag{2}$$

$$\left(x - \frac{w}{2} > 0\right) \wedge \left(x + \frac{w}{2} < W\right) \wedge \left(y - \frac{h}{2} > 0\right) \wedge \left(y + \frac{h}{2} < H\right) \tag{3}$$

where the symbol $\wedge$ represents the logical operator "AND." The x and y denote the coordinates of the center point of the rectangular box, while w represents the width of the box and h represents the height. W and H refer to the width and height of the image, respectively. Considering the uniqueness of each type of data, we apply different constraints to different types of data. More specifically, the vehicle, storage tank, and ship are constrained by (1)–(3), respectively, with parameters set to $T_{\text{size}} = 0.5$ and $T_{\text{shape}} = 0.25$. Additionally, airplane adhere to the constraints of (1) and (2), with parameters set as $T_{\text{size}} = 0.95$ and $T_{\text{shape}} = 0.5$. After generating bounding-box annotations, the K-Means++ algorithm is applied to each category of extracted images. As shown in Fig. 5, within each category's label set, there are correct instances of different scenes or types, as well as instances of noise caused by factors such as image background, low-quality data, or special circumstances. Therefore, the K value for clustering is determined through data analysis. Based on these constraints, pseudo labels that do not conform to the expected objects can be identified and filtered out, thereby reducing misjudgments and the impact of noise. Some examples of pseudo labels generated by our proposed IPLG for initial training data are shown in Fig. 6.

C. Progressive Self-Training With Relabeling

Our PST-R framework aims to enhance detection accuracy by training object detectors on both labeled and unlabeled data. To effectively leverage the unlabeled images, we adopt a teacher–student training paradigm. In this paradigm, the student model is optimized using both the unlabeled data with pseudo labels and the initial training data with bounding-box annotations, while the teacher model is responsible for generating the pseudo labels. Similar to previous works [25], [37], our PST-R employs teacher and student models that utilize detection networks with identical configurations. In this article, we adopt YOLOv5x [38] as our baseline detection network.

Specifically, given the initial training data $D^l = \{(x_i^l, y_i^l)\}_{i=1}^{N_l}$ and unlabeled data $D^u = \{x_i^u\}_{i=1}^{N_u}$, where N_l and N_u denote

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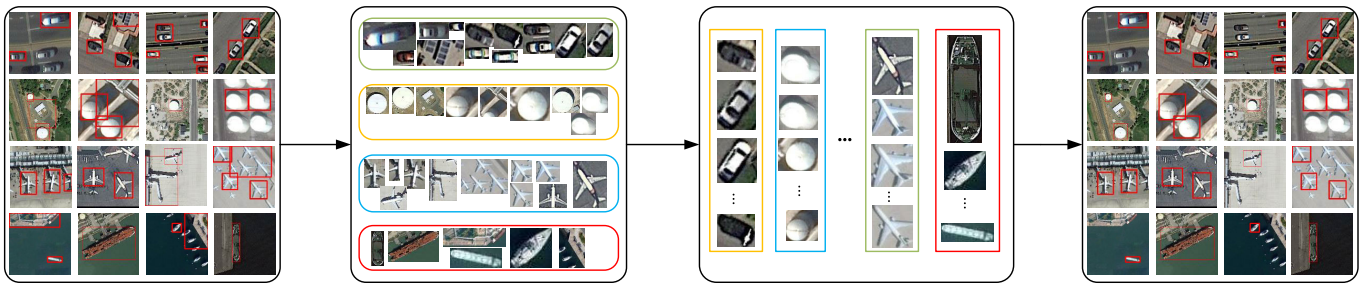


Fig. 5. Unsupervised clustering framework for filtering out noisy pseudo labels. After obtaining images with bounding-box labels using unsupervised object detection methods, we extract the object regions based on these bounding boxes. Subsequently, we perform unsupervised clustering on these extracted object images using K-Means++. Following this step, we discard the clustering folders that do not contain objects categorized as vehicles, storage tanks, airplanes, and ships. The remaining object images are considered reliable objects and are used for training the subsequent detection model.

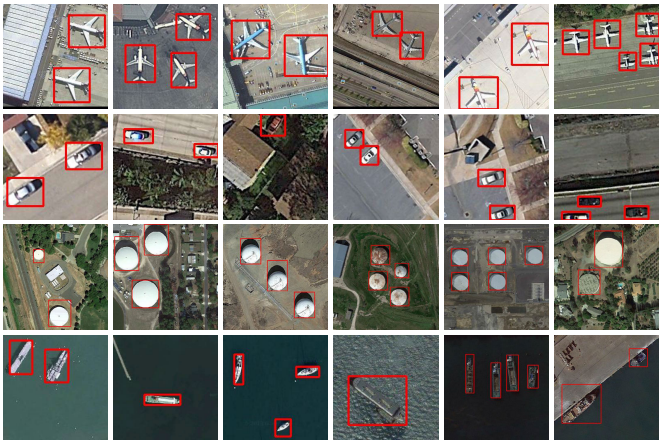


Fig. 6. Some examples of initial training samples with bounding-box annotations.

the number of labeled and unlabeled data. Each ground truth y_i^l consists of a set of classes c and bounding box coordinates b. We use the following steps to complete the self-training process: 1) train a detection model using the initial training data $D^l = \{(x_i^l, y_i^l)\}_{i=1}^{N_l}$; 2) infer the detection results for the unlabeled data and apply our proposed NPLFilter to filter out low-quality detection results; 3) combine initial training data with the purified pseudo labels and use them to train a new detection model; and 4) iterate the above steps to optimize the model and refine the pseudo labels.

D. Noisy Pseudo Label Filtering

To address the issue of model drift during progressive self-training, we introduce NPLFilter. NPLFilter effectively purifies generated pseudo labels by seamlessly integrating geometric constraints, prior knowledge, and category-adaptive thresholds. In the context of remote sensing imagery, each scene corresponds to specific object instances, and these instances are expected to adhere to objective patterns. For example, an airport may typically contain airplanes and vehicles, while the occurrence of ships is extremely rare. Consequently, we propose a noisy label filtering strategy based on prior knowledge, which aims to filter out pseudo labels that do not conform to objective patterns. Besides, some methods [22], [39] commonly employ manual setting of confidence score thresholds to filter out noisy pseudo boxes. However, such manual threshold setting is influenced by multiple interfering factors, and assigning the same threshold to different object categories neglects their individual attribute characteristics. In contrast to these methods, we introduce a category-adaptive threshold filtering approach that dynamically sets thresholds based on different object categories.

1) Geometric Constraint: Same as Section III-B4, the size information of the object is first used to filter out noisy pseudo labels. Specifically, we filter noisy pseudo labels by (1). Although a fixed T_{size} in (1) may lack universality due to diverse image dimensions, our observations indicate its efficacy in filtering out the majority of inaccurate pseudo labels. By excluding pseudo labels corresponding to objects with implausible sizes, the impact of noise can be effectively mitigated, thereby improving the quality of the labels.

2) Category-Adaptive Threshold: Upon completing the noisy pseudo label filtering based on the geometric constraint, we subsequently apply our proposed category-adaptive threshold filtering method to further refine the noisy pseudo labels. More specifically, we calculate the mean confidence score for each class of instances and utilize these means as thresholds for filtering out noise pseudo labels associated with the respective classes. The category-adaptive threshold can be represented as follows:

$$\mu_c = u_c - 0.2, u_c = \frac{1}{N_c} \sum_{i=1}^{N_c} s_i^c \tag{4}$$

where u_c denotes the average confidence score of all instances belonging to class c, N_c represents the total number of instances belonging to class c, and s_i^c is the confidence score of the ith instance belonging to class c. Note that due to noise in pseudo labels, the obtained u_c is higher than the actual value. Therefore, for each category c, we use u_c as a reference and choose a slightly lower threshold, namely the mean minus 0.2, as the final class-adaptive threshold to filter out noisy pseudo labels.

3) Prior Knowledge: To further enhance the quality of the remaining pseudo labels, we propose a strategy based on prior knowledge, which is executed through a series of specific steps. First, we count the number of instances of each category in the predicted image. The category (e.g., class c) with the highest occurrence is regarded as the most probable scene for this image. Then, within the category of the most probable scenario, we calculate the average confidence score of instances belonging to that category in the image and set the

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TABLE I
UNLABELED DATASETS IN THE EXPERIMENTS

Datasets	NWPU VHR-10.v2	DIOR	RSOD	LEVIR	UCAS-AOD	DOTA-v1.0
Number of Images	590	11,725	976	22,000	1,510	60,096

score lower than this average as the average to get more labels. The confidence score of modifying the target after applying the constraint is represented as follows:

$$S_j^c = \begin{cases} s_j^c, & s_j^c > \dfrac{1}{N_I^c}\displaystyle\sum_{j=1}^{N_I^c} s_j^c \\ \dfrac{1}{N_I^c}\displaystyle\sum_{j=1}^{N_I^c} s_j^c, & s_j^c \le \dfrac{1}{N_I^c}\displaystyle\sum_{j=1}^{N_I} s_j^c \end{cases} \tag{5}$$

where N_I^c represents the number of instances of class c in the image I and s_j^c is the confidence score of the jth instance of class c. Any instance of the cth class with a confidence score S_j^c less than μ_c is discarded and its annotations are removed. In this way, pseudo labels with low confidence scores for the corresponding categories in the scene to which the image may belong can be retained.

To filter out pseudo labels that do not conform to the objective pattern, we use statistics to obtain the semantic correlation between categories, which is calculated as follows:

$$R_c^o = \frac{\text{Count}(I_c \cap I_o)}{\text{Count}(I_c \bigcup I_o)}, \quad c, o \in C, I_c, I_o \subseteq T \tag{6}$$

where I_c and I_o, respectively, represent the set of images containing objects belonging to class c and o, $Count()$ denotes a counting function, C signifies the set of all object categories, and T refers to the dataset obtained after the inference of the previously trained model in the previous round of training. The specific implementation steps of the target semantic correlation constraint are as follows. First, the category with the largest number of instances in the image is counted, and then the semantic correlation between it and other categories is queried. Set ξ as the threshold for semantic correlation. Objects that are greater than or equal to ξ are considered related objects, while those below the threshold are considered unrelated objects. Finally, by introducing a constraint factor λ, and multiplying it with the object's attribute probability, uncorrelated objects are suppressed. The confidence score associated with the modification of the object, after the application of the constraint, can be quantified as follows:

$$S_j^o = \begin{cases} s_j^o & R_c^o \ge \xi \\ \lambda s_j^o, & R_c^o < \xi \end{cases} \tag{7}$$

where λ is 0.8 and ξ is 0.05. Any instance of the oth class with a confidence score S_j^o less than μ_o (i.e., the average confidence score of all instances belonging to class o) is discarded and its annotations are removed.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Datasets and Evaluation Metrics

1) Evaluation Datasets: We evaluate our PSTDet on two challenging datasets: NWPU VHR-10.v2 and DIOR. The NWPU VHR-10.v2 dataset contains ten categories, totaling 650 images. According to the dataset division criteria employed in previous works [12], [20], [21], we randomly choose 60 images containing vehicles, storage tanks, airplanes, and ships as test data. The remaining 590 images are being used as unlabeled data. DIOR is a large-scale and challenging remote sensing image object detection dataset, containing 20 common object categories. It comprises 23 463 images of size 800 × 800 pixels, including 5862 training images, 5863 validation images, and 11 738 test images. In our experiment, we treat the training and validation sets of the DIOR dataset as unlabeled data. To evaluate the performance of our proposed method, we utilize a subset of the DIOR test set consisting of 5400 images that contain vehicles, storage tanks, airplanes, and ships.

2) Unlabeled Datasets: We obtain unlabeled images from various sources, including RSOD [33], LEVIR [34], DOTA-v1.0 [40], NWPU VHR-10.v2 trainval set [16], DIOR trainval set [17], and UCAS-AOD [35]. Although these images are labeled, we treated them as unlabeled data. Detailed information about the constructed unlabeled dataset is presented in Table I. Importantly, in each round of inference, we currently utilize all unlabeled data collectively to generate pseudo boxes for subsequent training rounds. The RSOD dataset is comprised of 976 images across four object categories: airplane, storage tank, playground, and overpass. LEVIR contains 22 000 images covering three object categories: airplane, ship, and storage tank. UCAS-AOD consists of two categories: airplane and vehicle, with 1000 airplane images containing 7482 instances and 510 vehicle images containing 7114 instances. The DOTA-v1.0 dataset comprises 2806 images ranging in size from 800 × 800 to 4000 × 4000. To achieve precise detection, the images are cropped into 640 × 640 patches with a stride of 128. After cropping, it contains a total of 60 096 images.

3) Metrics: Consistent with standard practices in WSOD methods, we employ two critical metrics to evaluate the performance of our model on the testing set: the mean average precision (mAP) and the correct localization (CorLoc). The abovementioned two metrics treat each detection result as a positive detection when the IoU between the ground truth and bounding box is greater than 0.5.

B. Implementation Details

All experiments are implemented by PyTorch on a single RTX 2080 Ti GPU. The stochastic gradient descent (SGD) is set to 0.01, while the momentum and weight decay are set to 0.937 and 0.0005, respectively. The training epoch is set to 50 and we employ the mosaic data augmentation to enhance the robustness of the model.

C. Ablation Studies

To demonstrate the effectiveness of each component in our method, we conducted a series of ablation studies on the NWPU VHR-10.v2 and DIOR datasets.

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TABLE II

AP PERFORMANCE ON DIOR AND NWPU VHR-10.v2 TEST SET. THE FIRST ROUND OF TRAINING FOR ALL METHODS DIRECTLY USES INITIAL TRAINING IMAGES (i.e., 3999 IMAGES WITH BOUNDING-BOX ANNOTATIONS) AS THE TRAINING DATA. NOTE THAT THE w/0.7 FILTER INDICATES THAT THE CONFIDENCE THRESHOLD IS USED TO FILTER NOISE PSEUDO LABELS. HERE, WE SET THE CONFIDENCE THRESHOLD TO 0.7

Methods	Iteration	Train		Val		NWPU VHR-10.v2					DIOR				
		images	instances	images	instances	Vehicle	Storage tank	Aeroplane	Ship	mAP	Vehicle	Storage tank	Aeroplane	Ship	mAP
w/o Filter	iter1	3999	5254	997	1282	0.0917	0.191	0.318	0.247	0.212	0.136	0.130	0.231	0.0191	0.129
	iter2	22633	30639	2990	4180	0.202	0.338	0.652	0.403	0.399	0.177	0.174	0.386	0.0203	0.19
	iter3	35487	45775	3954	5347	0.146	0.591	0.788	0.407	0.483	0.142	0.233	0.463	0.0121	0.213
	iter4	52493	69289	4646	6528	0.229	0.776	0.823	0.427	0.564	0.193	0.283	0.520	0.0197	0.254
	iter5	71719	98602	5802	8503	0.201	0.821	0.852	0.430	0.576	0.195	0.325	0.563	0.0182	0.275
	iter6	79524	117621	6298	10230	0.233	0.847	0.972	0.435	0.622	0.201	0.376	0.645	0.016	0.31
	iter7	82512	130949	6351	10975	0.299	0.879	0.993	0.438	0.652	0.221	0.406	0.687	0.0185	0.333
	iter8	87558	149116	6608	12267	0.364	0.869	0.995	0.444	0.668	0.224	0.412	0.734	0.0175	0.347
	iter9	89568	163635	6679	12965	0.366	0.889	0.977	0.435	0.667	0.226	0.439	0.836	0.0162	0.379
	iter10	90185	173570	6635	12969	0.479	0.893	0.995	0.393	0.690	0.231	0.468	0.878	0.0174	0.399
	iter11	90956	192479	6663	13790	0.531	0.922	0.995	0.416	0.716	0.242	0.506	0.907	0.0181	0.418
	iter12	91226	211048	6644	14295	0.477	0.924	0.876	0.395	0.668	0.255	0.518	0.920	0.020	0.428
	iter13	91529	236708	6656	15437	0.634	0.923	0.995	0.417	0.742	0.255	0.534	0.943	0.0211	0.438
	iter14	91876	265077	6695	16749	0.659	0.926	0.977	0.468	0.758	0.260	0.526	0.949	0.0217	0.439
	iter15	91593	279101	6640	17553	0.731	0.938	0.995	0.446	0.777	0.273	0.509	0.955	0.0245	0.440
	iter16	92590	314804	6735	19430	0.729	0.933	0.995	0.473	0.782	0.266	0.520	0.946	0.0237	0.439
	iter17	92254	325540	6699	20026	0.641	0.916	0.89	0.505	0.738	0.266	0.511	0.946	0.0241	0.437
w/ 0.7 Filter	iter1	3999	5254	997	1282	0.0917	0.191	0.318	0.247	0.212	0.136	0.130	0.231	0.0191	0.129
	iter2	11835	15217	2110	2878	0.119	0.316	0.508	0.281	0.306	0.202	0.155	0.308	0.0201	0.171
	iter3	26847	31053	3333	4099	0.222	0.541	0.745	0.393	0.475	0.170	0.162	0.459	0.0387	0.207
	iter4	37212	39934	3467	4053	0.07	0.0926	0.723	0.352	0.309	0.120	0.124	0.471	0.0173	0.183
	iter5	54925	58017	4519	5198	0.105	0.0191	0.809	0.328	0.315	0.122	0.127	0.489	0.204	0.190
	iter6	64964	67974	5016	5674	0.103	0.0322	0.699	0.322	0.289	0.105	0.123	0.490	0.019	0.185
	iter7	75009	78101	5548	6197	0.0814	0.0115	0.691	0.304	0.272	0.0847	0.113	0.471	0.0172	0.171
NPLFilter	iter1	3999	5254	997	1282	0.0917	0.191	0.318	0.247	0.212	0.136	0.130	0.231	0.0191	0.129
	iter2	14004	20002	2384	3427	0.141	0.439	0.654	0.553	0.447	0.198	0.187	0.401	0.0195	0.201
	iter3	17284	25288	2890	4182	0.230	0.521	0.902	0.686	0.585	0.159	0.233	0.578	0.025	0.249
	iter4	21492	35509	3079	5010	0.229	0.654	0.918	0.756	0.639	0.198	0.222	0.622	0.0268	0.267
	iter5	28520	52270	3557	6135	0.332	0.765	0.946	0.72	0.691	0.220	0.259	0.665	0.0304	0.294
	iter6	33320	65497	4041	7370	0.328	0.811	0.967	0.737	0.711	0.212	0.269	0.765	0.0272	0.319
	iter7	41128	87110	4510	8850	0.400	0.860	0.990	0.737	0.747	0.237	0.296	0.813	0.0337	0.345
	iter8	42546	96927	4619	9510	0.435	0.887	0.991	0.696	0.753	0.225	0.441	0.807	0.022	0.374
	iter9	42982	101649	4665	9575	0.534	0.941	0.977	0.722	0.794	0.254	0.551	0.870	0.0255	0.425
	iter10	47465	118821	4807	10212	0.603	0.958	0.977	0.744	0.821	0.268	0.580	0.905	0.0285	0.445
	iter11	48100	127900	4916	10732	0.649	0.953	0.995	0.718	0.829	0.275	0.584	0.92	0.033	0.454
	iter12	49510	137784	5096	11399	0.664	0.936	0.995	0.704	0.825	0.281	0.630	0.931	0.0326	0.469
	iter13	51011	150212	5152	12000	0.736	0.923	0.995	0.717	0.843	0.291	0.644	0.947	0.0288	0.478
	iter14	56282	172292	5570	13599	0.708	0.895	0.995	0.742	0.835	0.281	0.661	0.95	0.0325	0.481
	iter15	58824	198529	5746	14606	0.744	0.924	0.995	0.743	**0.851**	0.303	0.666	0.961	0.0331	**0.491**
	iter16	55535	174353	5600	13540	0.697	0.904	0.895	0.729	0.806	0.301	0.672	0.963	0.0281	0.491

TABLE III

STUDIES OF THE COMPONENTS OF THE PROPOSED PSTDET

Filter Method	w/o Filter		✓	
	NPLFilter	✓		
Learning Stategy	w/o PST-R			✓
	PST-R	✓	✓	
NWPU VHR-10.v2	mAP	0.851	0.782	0.212
DIOR	mAP	0.491	0.439	0.129

1) Analysis of Components: In this section, we further evaluate the contribution of the two key components in our proposed PST-R: the NPLFilter and the progressive self-training mechanism. To this end, two baseline experiments are constructed. The first baseline experiment involves the PST-R without filtering noisy pseudo labels, aiming to validate the effectiveness of our proposed NPLFilter. As for the second baseline experiment, we will omit iterative learning as a control experiment to assess the impact of progressive training on the model.

Table III presents the detection results in terms of mAP for two baseline experiments. Upon analysis, we can discern the following: 1) compared to methods without noisy pseudo label filtering, our designed NPLFilter achieves mAP improvements of 6.9% and 5.2% on the NWPU VHR-10.v2 and DIOR datasets, respectively and 2) in contrast to approaches that do not utilize iterative learning, our progressive self-training mechanism significantly enhances the mAP by 63.9% and 36.2% on the NWPU VHR-10.v2 and DIOR datasets, respectively. These findings unequivocally highlight the efficacy of the NPLFilter and underscore the crucial role of the progressive self-training mechanism in advancing object detection performance. One potential contributing factor to this improvement lies in the progressive self-training approach, which harnesses joint optimization of initial training data and unlabeled target domain data to facilitate mutual enhancement between the model and pseudo labels throughout the iterative learning process. This, in turn, bolsters the model's robustness and alleviates domain variances between the initial training data and the target domain data.

a) Effectiveness of pseudo label filtering: To further validate the impact of NPLFilter on model performance, we evaluated the performance of three pseudo label filtering approaches: no noisy pseudo label filtering (w/o Filter), filtering noisy pseudo labels based on confidence threshold, and our proposed NPLFilter. The results are illustrated in Table II.

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TABLE IV
AP COMPARISON RESULTS OF YOLOV4 AND YOLOV8X REPLACING YOLOV5X FOR OBJECT DETECTION ON THE DIOR DATASET

Detector	Iteration	Vehicle	Storage tank	Aeroplane	Ship	mAP
YOLOv4	iter1	0.061	0.124	0.248	0.008	0.110
YOLOv4	iter2	0.083	0.194	0.459	0.008	0.186
YOLOv5x	iter1	0.136	0.130	0.231	0.0191	0.129
YOLOv5x	iter2	0.198	0.187	0.401	0.0195	0.201
YOLOv8x	iter1	0.113	0.131	0.260	0.0179	0.130
YOLOv8x	iter2	0.413	0.262	0.373	0.0264	0.269

From these comparison results, one can easily observe that our method achieves better performance than other competitors. The reason may come from the following two aspects: 1) our NPLFilter leverages the geometric constraint and prior knowledge to effectively filter noisy pseudo labels by considering target characteristics and 2) applying a confidence threshold-based filtering method (w/ 0.7) with a uniform threshold across different object classes overlooks the distinctive characteristics of each class, which can hinder certain classes from achieving optimal performance. Moreover, employing this method to eliminate noisy pseudo labels might inadvertently discard valuable data, leading to a substantial reduction in training data volume and subsequently diminishing model accuracy. To address this limitation, our NPLFilter adopts a category-AF strategy that takes into account the unique properties of each class, thereby mitigating the impact of noisy pseudo labels at the model level.

b) Analyzing the impact of detectors: Assessing the performance of PSTDet with widely used object detectors would provide valuable insights. In our experiment, we use YOLOv4 [41] and YOLOv8x instead of YOLOv5x to complete the comparative experiment under the same training strategy. The results, as presented in Table IV, indicate that integrating PSTDet with a more robust detector leads to a significant improvement in detection accuracy. These experiments demonstrate that our method is not limited by the choice of detector, highlighting the versatility and potential of our proposed approach for real-world applications.

c) Select the number of iterations: In general, more iterative training tends to optimize the model better, leading to improved predictive outcomes. However, there is no mechanism to guarantee the convergence of the iteration. The strategy adopted in this article is to halt the iterative process when the number of instances in the training images ceases to increase. It is important to highlight that apart from the initial round where 997 images from the initial training data are used, the subsequent validation sets exclusively consist of images from the DIOR validation set. Furthermore, the bounding-box annotations for these images are inferred from the previous round. Table II presents the influence of different iteration numbers T using various filtering methods on model performance. Experimental results indicate that as T increases, performance initially improves, reaching a peak before exhibiting oscillations. This outcome suggests that progress self-training is effective for further enhancing detection performance. Fig. 9 shows some detection results of each iteration of our proposed approach.

TABLE V
PERFORMANCE COMPARISONS OF DIFFERENT METHODS IN TERMS OF AP VALUES ON THE NWPU-VHR-10.V2 TEST SET

Method	Aeroplane	Ship	Vehicle	Storage tank	mAP
Fast-RCNN [42]	0.9091	0.9060	0.6984	0.8929	0.8516
Faster-RCNN [43]	0.9090	0.8630	0.7814	0.9053	0.8647
RICNN [44]	0.8871	0.7834	0.7201	0.8633	0.8135
RCNN [45]	0.8537	0.8888	0.4992	0.6278	0.7174
Transfer CNN [46]	0.6603	0.5713	0.4127	0.8501	0.6236
COPD [47]	0.6225	0.6937	0.4428	0.6452	0.6011
RICO [16]	0.9970	0.9080	0.8714	0.9061	0.9206
YOLOv5x [38]	0.9950	0.9240	0.9340	0.4490	0.8255
WSDDN [48]	0.3008	0.4172	0.0360	0.3498	0.2760
OICR [49]	0.1366	0.6735	0.0373	0.5716	0.3548
PCL [50]	0.2600	0.6376	0.1567	0.0250	0.2766
MELM [51]	0.8086	0.6930	0.0868	0.1048	0.4233
DCL [21]	0.7270	0.7425	0.3500	0.3705	0.5475
TCANet [20]	0.8943	0.7818	0.2830	0.7842	0.6858
DPLG [52]	0.8090	0.7830	0.0870	0.1050	0.4460
OS [12]	0.8164	0.6833	0.2587	0.6531	0.6029
SPG+OICR [11]	0.4280	0.7256	0.2475	0.5882	0.4973
SPG+MELM [11]	0.9042	0.8100	0.4299	0.5953	0.6849
Qian *et al.* [53]	0.8760	0.8010	0.3560	0.5730	0.6215
Tan *et al.* [14]	0.9840	0.8600	0.4010	0.7120	0.7393
AE-IS [54]	0.9100	**0.8820**	0.4580	0.7830	0.7583
Ours	**0.9950**	0.7430	**0.7440**	**0.9240**	**0.8510**

D. Comparison With State-of-the-Arts

To thoroughly evaluate the effectiveness of our proposed method, we conduct an extensive comparison against state-of-the-art techniques in fully supervised, weakly supervised, semi-supervised, and zero-shot object detection (ZSD). This evaluation encompasses both the NWPU VHR-10.v2 and DIOR datasets, ensuring a comprehensive analysis across diverse benchmarks.

1) Experimental Results on the NWPU VHR-10.v2 Dataset: We conduct a comprehensive evaluation of our method on the NWPU VHR-10.v2 dataset. The results are shown in Tables V and VI. The experimental results demonstrate that our PSTDet performs remarkably well, achieving the state-of-the-art mAP score of 85.10% and 99.55% CorLoc, significantly outperforming all weakly supervised models. Specifically, our PSTDet method demonstrates superior performance compared to WSOD methods such as WSDDN, OICR, PCL, MELM, DCL, TCANet, DPLG, OS, SPG + OICR, SPG + MELM, Qian et al., Tan et al., and AE-IS, achieving improvements in mAP values on the NWPU VHR-10.v2 dataset by margins of 57.50%, 49.62%, 57.44%, 42.77%, 30.35%, 16.52%, 40.5%, 24.81%, 35.37%, 16.61%, 22.95%, 11.17%, and 9.27%, respectively. Moreover, our method surpasses FSOD methods, including RICNN, RCNN, Transfer CNN, COPD, and YOLOv5x, by margins of 3.75%, 13.36%, 22.74%, 24.99%, and 2.55%, respectively.

2) Experimental Results on the DIOR Dataset: We compared the performance of PSTDet against 15 state-of-the-art methods on the DIOR dataset. The results presented in Table VII indicate that our PSTDet outperforms all other weakly supervised methods, achieving a mAP of 49.10% on the test set of the DIOR dataset. Besides, the CorLoc results, as shown in Table VIII, further reveal the effectiveness of our proposed PSTDet. Our results demonstrate the potential for PSTDet to be utilized in real-world scenarios where resources or time constraints may make traditional fully supervised methods unfeasible. By achieving state-of-the-art performance

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TABLE VI
PERFORMANCE COMPARISONS (CORLOC) AMONG DIFFERENT METHODS ON THE NWPU-VHR-10.V2 TRAINVAL SET

Method	Aeroplane	Ship	Vehicle	Storage tank	CorLoc
WSDDN [48]	0.2232	0.3681	0.0289	0.3995	0.2549
OICR [49]	0.2941	0.8333	0.1444	0.2051	0.3692
PCL [50]	0.1176	0.5000	0.1556	0.1282	0.2254
MELM [51]	0.8596	0.7742	0.1463	0.2143	0.4986
TCANet [20]	0.9691	0.9178	0.5556	0.9513	0.8485
SPG+MELM [11]	0.981	0.927	0.600	0.701	0.8023
Qian *et al.* [53]	0.944	0.866	0.566	0.685	0.7653
Tan *et al.* [14]	1.000	0.951	0.533	0.942	0.8565
AE-IS [54]	0.983	0.942	0.458	0.783	0.7915
Ours	**1.000**	**1.000**	**0.9820**	**1.000**	**0.9955**

TABLE VII
PERFORMANCE COMPARISONS OF DIFFERENT METHODS IN TERMS OF AP VALUES ON THE DIOR TEST SET

Method	Aeroplane	Ship	Vehicle	Storage tank	mAP
Fast-RCNN [42]	0.4417	0.3213	0.1920	0.3504	0.3264
Faster-RCNN [43]	0.5028	0.5848	0.3477	0.4235	0.4647
YOLOv5x [38]	0.8300	0.8970	0.6210	0.8090	0.7893
WSDDN [48]	0.0906	0.0070	0.0455	0.0395	0.0457
OICR [49]	0.0870	0.0909	0.0909	0.0710	0.0850
PCL [50]	0.2152	0.0909	0.0455	0.0909	0.1106
MELM [51]	0.2814	0.0909	0.0004	0.1502	0.1307
DCL [21]	0.2089	0.0909	0.0727	0.0909	0.1159
TCANet [20]	0.2513	0.0909	0.0909	0.1374	0.1426
OS [12]	0.4110	**0.2134**	0.1834	0.2466	0.2636
SPG+OICR [11]	0.1280	0.0923	0.0912	0.0119	0.0809
SPG+MELM [11]	0.3132	0.1002	0.0315	0.1510	0.1490
Qian *et al.* [53]	0.2910	0.1850	0.1130	0.1170	0.1765
Tan *et al.* [14]	0.4290	0.0910	0.0630	0.3060	0.2223
AE-IS [54]	0.3180	0.1120	0.0800	0.1630	0.1683
Ours	**0.9610**	0.0331	**0.3030**	**0.6660**	**0.4910**

TABLE VIII
PERFORMANCE COMPARISONS (CORLOC) AMONG DIFFERENT METHODS ON THE DIOR TRAINVAL SET

Method	Aeroplane	Ship	Vehicle	Storage tank	CorLoc
WSDDN [48]	0.0572	0.2296	0.1156	0.7961	0.2996
OICR [49]	0.1598	0.3180	0.1583	0.8129	0.3623
PCL [50]	0.6114	0.3794	0.1050	0.8065	0.4756
MELM [51]	0.7698	0.2819	0.0462	0.8297	0.4819
TCANet [20]	0.8158	0.3063	0.1045	0.9146	0.5353
SPG+MELM [11]	0.805	0.352	0.209	0.900	0.5665
Qian *et al.* [53]	0.855	0.397	0.209	0.770	0.5578
Tan *et al.* [14]	0.960	0.386	0.302	0.903	0.6378
AE-IS [54]	0.914	0.402	0.123	0.915	0.5885
Ours	**0.985**	**0.658**	**0.730**	**0.977**	**0.838**

with weak supervision, our approach offers a practical solution for many challenging real-world detection tasks.

3) Comparison With SSOD Methods: Our PSTDet aligns with the core tenets of SSOD approaches, which leverage a limited number of labeled images alongside a substantial reservoir of unlabeled images to bolster performance. Diverging from these methodologies, ours eschews the necessity for manually annotated bounding-box annotations, instead automatically generating precise bounding-box annotations through a fusion of unsupervised image clustering, unsupervised instance detection, and geometric constraints. Subsequently, detection knowledge is acquired through learning from this data corpus. Within this section, we compare our method on the DIOR test set with various competitors: Efficient Teacher [55], Dense Teacher [56], and Soft Teacher [27]. Specifically, we employ the training set from the initial training data as the source of labeled images, the training set of the DIOR dataset as unlabeled images, the validation set of the initial training data as the validation set, and the DIOR test set as our evaluation benchmark across all experimental setups. As delineated in Table IX, our methodology achieves preeminence, exhibiting a substantial lead over its competitors. The rationale for this superiority can be encapsulated as follows: 1) SSOD methods necessitate labeled images of superior quality that mirror the domain distribution of unlabeled images, thereby confronting challenges in attaining satisfactory performance when confronted with disparate domain distributions and 2) our PSTDet method adopts an iterative learning scheme to gradually enhance the performance of the segmentation model. It is noteworthy that all referenced methodologies adhere to their respective publications' methodologies and are implemented in PyTorch.

TABLE IX
COMPARISONS WITH STATE-OF-THE-ART SSOD METHODS ON THE DIOR TEST SET

Method	Aeroplane	Ship	Vehicle	Storage tank	mAP
Efficient Teacher [55]	0.815	0.0251	0.0784	0.253	0.293
Dense Teacher [56]	0.337	0.019	0.056	0.025	0.109
Soft Teacher [27]	0.386	0.025	0.080	0.010	0.125
Ours	**0.961**	**0.0331**	**0.303**	**0.666**	**0.491**

4) Comparison With ZSD Methods: ZSD endeavors to identify and localize "unseen" object instances solely based on their semantic information. Initially, the model is trained to establish associations between visual and semantic domains for known objects. Subsequently, it leverages this learned knowledge to detect entirely unfamiliar objects. In this article, we compare our proposed method with several existing ZSD techniques, including PL [57], Hayat et al. [58], and RRFS [59]. We conduct a detailed comparison between our proposed approach and these methods on both the DIOR and NWPU VHR-10 v2 datasets. In our experiments, we utilized the same vocabulary and semantic data as described in PL [57], Hayat et al. [58], and RRFS [59]. Specifically, we selected three categories: airplane, ship, and vehicle, with airplane and the ship being seen classes while vehicle served as the unseen class. Table X presents the experimental results obtained from the DIOR dataset and NWPU VHR-10.v2 dataset, respectively. Our method demonstrates superior performance compared to existing approaches. This can be attributed to IPLG, which establishes a solid foundation for high-performance models by providing accurate initial bounding-box annotations. Additionally, PST-R enhances the reliability of pseudo labels through progressive self-training, while NPLFilter optimizes the pseudo labels to improve model accuracy. Unlike our method, existing ZSD techniques often rely on expert knowledge to craft extensive text prompts and may suffer from misalignment between high-level semantic features from text and low-level features from image patches in object detection.

E. Qualitative Results

Fig. 7 presents the detection results of our method on the NWPU VHR-10.v2 and DIOR datasets test sets, as well as on unlabeled images. The first four columns of the figure showcase the detection outcomes on the test sets of NWPU

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Fig. 7. Qualitative detection results on the NWPU VHR-10.v2, DIOR, and unlabeled datasets.



Fig. 8. Some failure cases. (a) Few missed objects. (b) Few false positives.

TABLE X

CLASS-WISE AP ON SEEN AND UNSEEN CLASSES OF NWPU VHR-10.V2 AND DIOR DATASETS FOR ZSD

Datasets	Methods	Seen		Unseen
		Aeroplane	Ship	Vehicle
NWPU VHR-10.v2	PL [57]	0.713	0.614	0.011
	Hayat *et al.* [58]	1.000	0.838	0.144
	RRFS [59]	0.970	0.371	0.180
	Ours	0.995	0.743	0.744
DIOR	PL [57]	0.252	0.407	0.001
	Hayat *et al.* [58]	0.638	0.743	0.033
	RRFS [59]	0.599	0.599	0.025
	Ours	0.961	0.033	0.303

VHR-10.v2 and DIOR datasets, while columns five through eight illustrate the results on the unlabeled images. Notably, our PSTDet method exhibits exceptional performance by successfully detecting small ships and airplanes. This accomplishment highlights the efficacy of our proposed approach in addressing the inherent challenges associated with remote sensing object detection.

Although our approach outperforms other state-of-the-art WSOD methods in terms of performance, it also exhibits failed detection results in the following scenarios.

1) *Dense Object Misdetection:* When objects are in close proximity to each other, the collective features of multiple adjacent objects of the same category are often more discriminative than the features of individual objects. Consequently, multiple neighboring objects may be localized within a single large bounding box, as shown in Fig. 8(a).
2) *Similar Characteristic Misdetection:* Certain foreground objects have similar features to other foreground objects or background regions, leading to potential interference in their detection, as shown in Fig. 8(b).

In the future, we will consider introducing effective strategies to alleviate the aforementioned challenges.

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Fig. 9. Object detection results as iterations increase.

F. Computational Complexity

The computational complexity of our method relies on two crucial factors: the number of iterations and the complexity of the basic models, such as YOLOv5x, employed within each iteration. In this study, we implemented an iterative training strategy that involved stopping the training process after the 15th iteration, with each iteration comprising around 50 epochs. For example, in the second iteration, we trained the YOLOv5x model for 50 epochs using an NVIDIA RTX 2080 Ti GPU. Each epoch took approximately 803.58 s, resulting in a total training time of approximately 11.1 h. Following the training process, we evaluated the performance of the trained model on two distinct datasets, one containing 5400 images and the other containing 60 images. Notably, the evaluation process consumed only 184.31 and 17.4 s for the larger and smaller datasets, respectively. To further enhance the model's performance, we utilized the trained model to perform inference on unlabeled data consisting of 96 897 images. This allowed us to obtain data with pseudo labels, which were subsequently used to refine the model's performance. The detection process for this large dataset took approximately 58.68 min. These results highlight the efficiency of our approach, as the training time pales in comparison to the time-consuming nature of manual data annotation.

V. Future Direction

Recently, foundation models using self-supervised learning have attracted great attention due to their potential to revolutionize the field of visual representation learning. In this domain, Hong et al. [60] proposed the first universal remote sensing foundation model, SpectralGPT, specifically designed for processing spectral remote sensing images using a novel 3-D generative pretrained transformer (GPT). After being trained on one million spectral RS images, the SpectralGPT model has over 600 million parameters and achieves state-of-the-art performance in four downstream tasks such as classification, segmentation, and change detection. Mall et al. [61] designed an innovative method that can train a remote sensing image visual language model without the need for any textual annotations. Their research achievements have exceled in multiple key tasks, covering zero-shot learning, open-vocabulary image classification, retrieval, segmentation, and visual question answering domains. Furthermore, Hong et al. [62] have developed an advanced deep neural network architecture called HighDAN. By integrating adversarial learning-based domain adaptation principles into HR-Net with Dice Loss, they have significantly pushed the boundaries of semantic segmentation accuracy and generalization capabilities in cross-city studies. These research efforts are of paramount significance for advancing the intelligent understanding of remote sensing imagery. Moreover, they hold strong practical implications for bridging the gap between theory and real-world applications in the industry. Through the continuous development and refinement of these foundational models, we believe that fully automated remote sensing interpretation will be achieved shortly, benefiting users in various domains such as environmental monitoring and disaster response. In the future, we plan to utilize PSTDet to provide labeled data for training foundational remote sensing models and further research effective methods for fine-tuning these models, thereby contributing to the advancement of intelligent understanding of remote sensing images.

VI. Conclusion

In this study, we present PSTDet, a novel framework for object detection in remote sensing imagery that eliminates the need for manually labeled data. PSTDet leverages the IPLG to automatically generate high-quality pseudo labels for the initial training data. The PST-R is then employed to progressively learn detection knowledge using both the initial training data and the available unlabeled data. Additionally, we introduce the NPLFilter to purify the quality of the pseudo labels during the progressive self-training process. To evaluate the performance of our proposed method, extensive experiments are conducted on the NWPU VHR-10.v2 and DIOR datasets. The results demonstrate the effectiveness and superiority of PSTDet in remote sensing object detection.

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